Chapter 4 (Continued): Laboratory Experimental Summary and Research Protocol

In summary, this research prototype explores the practical boundaries of content-stream steganography in PDF documents. By combining strict cryptographic authentication, rigorous visual fidelity metrics, and comprehensive transformation benchmarks, we establish an empirical baseline for covert recipient attribution.

Students and researchers analyzing this system should observe that cryptographic verification and visual imperceptibility are complementary requirements. A watermark that is visually imperceptible but cryptographically malleable fails to provide non-repudiation. Conversely, a watermark that is cryptographically robust but visually disruptive undermines document usability.

Our test framework quantifies both dimensions through automated bit-error-rate analysis and pixel-level distortion calculations, ensuring all research conclusions are supported by reproducible empirical evidence.

Chapter 4 (Continued): Security Boundaries and Failure Modes

Every security mechanism must be evaluated within an explicit adversarial threat model. In our evaluation, we subject watermarked documents to seven distinct transformation categories: pristine baseline transmission, lossless optimization and deflation re-saving, catalog dictionary metadata tampering, extraneous content injection, page order permutation, margin cropping, and bitmap rasterization.

Lossless re-saving and catalog tampering leave the content stream kerning unaffected, allowing complete payload recovery. In contrast, adversarial operations that alter the relative sequence of text lines inevitably introduce bit synchronization errors. Most critically, rasterizing a document to an image removes all text operators entirely, causing the text-level carrier to be completely destroyed.

Understanding these boundaries prevents researchers from making unsubstantiated claims regarding steganographic invulnerability. Steganographic kerning modulation excels as a software-level vector carrier, but it is not intended to replace physical print-scan watermarking or robust image transforms.

Chapter 4 (Continued): Cryptographic Authentication and Capacity Modeling

Security in covert marking cannot rely on obscurity alone. Following Kerckhoffs principle, an adversary may understand the exact modulation algorithms, carrier structures, and stream parsing procedures. Consequently, authentication security must reside entirely in an uncompromised cryptographic key.

We utilize HMAC-SHA256 to bind document identifiers, recipient credentials, and fresh cryptographic nonces into an unforgeable 256-bit message authentication code. Truncating the tag to arbitrary low-bit lengths (e.g. 64 bits) compromises forgery resistance and enables brute-force collision attacks. Therefore, our architecture enforces full untruncated 256-bit tags, requiring a minimum payload capacity budget of approximately 664 to 700 bits per document.

If an input document contains insufficient word boundaries to host the complete authenticated payload, the encoder fails closed with an InsufficientCapacityError rather than silently discarding security margins. The capacity budget dictates that approximately sixty to seventy lines of standard text are necessary to embed the complete authentication frame.

Chapter 4 (Continued): Mathematical Formulations of Micro-Kerning

In the Portable Document Format, text is positioned using coordinate matrices and rendered via operators such as Tj and TJ. The TJ operator accepts an array of alternating string literals and horizontal glyph displacement numbers measured in thousandths of an em. A negative integer shifts the current text position to the right, slightly expanding the inter-word gap between successive tokens.

By defining a binary alphabet where bit zero corresponds to natural spacing and bit one corresponds to a minor displacement delta, an encoder can serialize arbitrary cryptographic payloads into the document layout. To ensure frame synchronization, the bitstream is prefixed with a distinctive preamble sequence, followed by version metadata, document identifiers, recipient credentials, a random cryptographic salt, a complete 256-bit HMAC tag, and an error-detecting checksum.

Because the string characters themselves remain untouched, standard text selection, copy-pasting, and search indexing operate with zero degradation. During forensic verification, an investigator inspects the content stream, decodes the sequence of displacement values, validates the checksum, and verifies the message authentication code against the authoritative master secret key. This dual-layered architecture provides defensive depth alongside traditional catalog and annotation metadata carriers.

COMPUTER SYSTEMS AND INFORMATION SECURITY
Chapter 4: Principles of Digital Watermarking and Covert Channels

Information hiding represents a vital domain within applied cryptography and digital rights management. Unlike standard encryption, which renders messages visibly scrambled, steganography aims to conceal the very existence of the communication. Within document processing systems, carriers can take several structural forms across vector and metadata representations.

Traditional techniques include invisible annotation layers, document metadata modifications, font substitution, and content-stream kerning adjustments. When designing a covert marking scheme, engineers must balance capacity, imperceptibility, and robustness. A scheme that embeds too many bits per line risks introducing perceptible layout distortions or reflow artifacts. Conversely, a scheme with insufficient capacity cannot carry a modern untruncated cryptographic authentication tag such as HMAC-SHA256, leaving the system vulnerable to bit-flipping attacks.

In this laboratory experiment, we evaluate whether subtle variations in inter-word kerning within PDF content streams can reliably convey unforgeable recipient attributions while maintaining zero visual divergence under normal reading. Every word boundary presents an opportunity to encode a binary digit by slightly displacing the character spacing. However, physical document transformations such as printing, scanning, or bitmap rasterization inherently strip the underlying vector text operators, delineating the fundamental physical boundary of text-level steganography.